

Week 3 - Part 2 - Osprey1st College Algebra - Summer 2024

Section covered: 2.4, 2.5, 3.1, and 3.2

Factoring Trinomials

Factoring Quadratic Functions with leading coefficient $a = 1$

$$x^2 + 7x + 12 = (x + \boxed{}) \cdot (x + \boxed{})$$

To determine the last term of each binomial factor, we look for two integers whose product is 12 and whose sum is 7. Since our numbers must have a positive product and a positive sum, we list pairs of positive integer factors of 12 only.

$$(x + 1)(x + 12), \quad (x + 2)(x + 6), \quad \text{or} \quad (x + 3)(x + 4)?$$

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Problem 9. Factor each trinomial.

(a) $x^2 + 16x + 15$

$$(x + 15)(x + 1)$$

$$\begin{array}{r} +15 \\ +1 \\ \hline +16 \end{array}$$

(b) $x^2 - 6x - 16$

$$(x - 8)(x + 2)$$

$$\begin{array}{r} -16 \\ +2 \\ \hline -6 \end{array}$$

(c) $x^2 - 11x + 18$

$$(x - 9)(x - 2)$$

$$\begin{array}{r} +18 \\ -9 \\ \hline -11 \end{array}$$

(d) $x^2 + 7x - 30$

$$(x + 10)(x - 3)$$

$$\begin{array}{r} -30 \\ +10 \\ \hline +7 \end{array}$$

$$\begin{array}{r} 30 \\ 1, 30 \\ 2, 15 \\ 3, 10 \\ 4, 7.5 \\ \hline 5, 6 \end{array}$$

(e) $x^2 + 12x + 36$

$$(x + 6)(x + 6)$$

$$\begin{array}{r} +36 \\ +6 \\ \hline +12 \end{array}$$

$$(x + 6)^2$$

(f) $2x^2 - 28x + 90$

$$2(x^2 - 14x + 45)$$

$$\begin{array}{r} +45 \\ -5 \\ \hline -14 \end{array}$$

$$2(x - 5)(x - 9)$$

(g) $2v^2 - 24v + 42$

$$2(v^2 - 12v + 21)$$

$$\begin{array}{r} +21 \\ +2 \\ \hline -12 \end{array}$$

$$\begin{array}{r} 21 \\ 1, 21 \\ 2, 10.5 \\ 3, 7 \\ \hline \end{array}$$

last sign positive
both signs are same
as the middle sign

last sign negative
signs are different
bigger # gets the middle sign

Problem 10. Factor the polynomials.

(a) $x^2 - 10x + 25$

$(x-5)(x-5)$ ~~$\begin{array}{r} +25 \\ -5 \quad -5 \\ -10 \end{array}$~~ $(x-5)^2$

(b) $2x^2 - 16x + 32$

$2(x^2 - 8x + 16)$ ~~$\begin{array}{r} +16 \\ -4 \quad -4 \\ -8 \end{array}$~~ $2(x-4)(x-4)$
 $2(x-4)^2$

Factoring Quadratic Functions with leading coefficient $a \neq 1$; ac method

$$5x^2 + 11x + 2$$

We look for two numbers that satisfy

$$\square \cdot \square = 5 \cdot 2$$

and

$$\square + \square = 11$$

These two numbers are 10 and 1.

$$5x^2 + \underbrace{10x + x}_{11x} + 2 = (5x^2 + 10x) + (x + 2) = 5x(x + 2) + x + 2 = (x + 2)(5x + 1)$$

Factoring Quadratic Functions with leading coefficient $a \neq 1$; "bottom-up" or "slide or divide" method

Factor out a greatest common factor, if applicable.

$$5x^2 + 11x + 2$$

We look for two numbers that satisfy

$$\square \cdot \square = 5 \cdot 2$$

and

$$\square + \square = 11$$

These two numbers are 10 and 1. Then we divide those numbers by $a = 5$ and simplify common factors:

$$\frac{10}{5} = \frac{2}{1} \quad \text{and} \quad \frac{1}{5}$$

We read the factorized form from the bottom up.

$$5x^2 + 11x + 2 = (1x + 2)(5x + 1) = (x + 2)(5x + 1)$$

Problem 11. Factor each trinomial.

(a) $3x^2 - 11x - 10$

$$(3x - 6)(3x + 5)$$

(3)

$$(x - 2)(3x + 5)$$

$$\begin{array}{r} 3x - 10 \\ -30 \\ -6 \quad +5 \\ \hline -1 \end{array}$$

(b) $4x^2 + 20x + 25$

$$(4x + 10)(4x + 10)$$

(2) (2)

$$(2x + 5)(2x + 5)$$

$$\begin{array}{r} 4x + 25 \\ +100 \\ +10 \quad +10 \\ \hline +20 \end{array}$$

(c) $6z^2 - 19z + 3$

$$(6z - 18)(6z - 1)$$

(6)

$$(z - 3)(6z - 1)$$

$$\begin{array}{r} 6z + 3 \\ +18 \\ -18 \quad -1 \\ \hline -19 \end{array}$$

(d) $5z^2 + 4z - 12$

(e) $6x^2 - 32x + 32$

Factor by grouping

$$(3x^2 - 6x)(+5x - 10)$$

$$\underline{\underline{3x(x - 2)}} + \underline{\underline{5(x - 2)}}$$

$$(x - 2)(3x + 5)$$

(f) $8z^2 - 22z + 9$

(g) $24x^2 + 42x + 15$

(h) $-3w^2 + 10w - 8$

Factoring Perfect Square Trinomials

$$A^2 + 2AB + B^2 = (A + B)^2$$

$$A^2 - 2AB + B^2 = (A - B)^2$$

Problem 12. Factor each trinomial.

(a) $49x^2 + 28xy + 4y^2$

(b) $w^2 + 18wv + 81v^2$

(c) $81a^2b^2 - 90ab + 25$

Factoring Binomials

Special Bionomials

Difference of Squares $A^2 - B^2 = (A + B)(A - B)$

Difference of Cubes $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$

Sum of Cubes $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$

Note: Sum of squares $A^2 + B^2$ is ”**prime**”, which means it cannot be factored.

Problem 13. Factor the binomials completely, if possible.

(a) $x^2 - 36$

(b) $25y^2 - 4$

(c) * $49 - z^2$

(d) $9x^2 + 1$

(e) * $80u^2 - 45$

(f) $16a^4 - 25b^2$

(g) * $w^3 - w^3x^4$

(h) $t^3 + 27$

(i) $x^3 - 8y^3$

(j) * $125 - 8x^3$

Radical Notation

The expression $a^{\frac{1}{n}} = \sqrt[n]{a}$, where n is a natural number, that is

n is **even**
 a is positive

$\sqrt[n]{a}$ is the positive real number
 whose n th power is a .
 For example $\sqrt[4]{16} = 2$ since $2 > 0$ and $2^4 = 16$

n is **odd**
 a is any real number

$\sqrt[n]{a}$ is the real number (positive or negative)
 whose n th power is a .
 For example $\sqrt[3]{-64} = -4$ since $(-4)^3 = -64$

For any natural number n , $\sqrt[n]{0} = 0$

Radical Expression:

Index **Radical Symbol**
 **Radicand**

Evaluating $\sqrt[n]{a^n}$

If n is even natural number, then $\sqrt[n]{a^n} = |a|$.

If n is odd natural number, then $\sqrt[n]{a^n} = a$.

For example $\sqrt[3]{5^3} = 5$, $\sqrt[4]{2^4} = 2$ but $\sqrt[4]{(-2)^4} = |-2| = 2$.

Problem 1. Simplify each expression, if possible.

(a) $\sqrt[3]{2^3}$

(e) $\sqrt{(-11)^2}$

(b) $\sqrt[4]{(-3)^4}$

(f) $\sqrt{(-6)^2}$

(c) $\sqrt[5]{(-6)^5}$

(g) $\sqrt[7]{x^7}$

(d) $-\sqrt{3^4}$

(h) $\sqrt[8]{z^8}$

$$(i) \sqrt{y^2}$$

$$(k) \text{ * } \sqrt[8]{(4-3z)^8}$$

$$(j) \text{ * } \sqrt[7]{(1-x)^7}$$

Rules for Radicals

Rules for Radicals

Suppose that a and b represent real numbers, and m and n represent positive integers for which the indicated roots are real numbers.

$$\text{Product Rule} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

$$\text{Quotient Rule} \quad \frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$$

$$\text{Power Rule} \quad \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

Problem 2. Simplify. Assume all variable expressions represent positive real numbers.

$$(a) \sqrt[3]{2} \cdot \sqrt[3]{4}$$

$$(d) \sqrt{25z}$$

$$(g) \frac{\sqrt{50x^3}}{\sqrt{2x}}$$

$$(b) \text{ * } \sqrt{14} \cdot \sqrt{2}$$

$$(e) \sqrt[4]{\frac{a^8}{b^4}}$$

$$(h) \text{ * } \sqrt{3b^2} \cdot \sqrt{6b^5}$$

$$(c) \sqrt[5]{n^3} \cdot \sqrt[5]{n^2}$$

$$(f) \text{ * } \sqrt{18x^4} \cdot \sqrt{2x^3}$$

Simplified Radicals

An expression with radicals is simplified when all of the following conditions are satisfied.

1. The radicand has no factor raised to a power greater than or equal to the index.
2. The radicand has no fractions.
3. No denominator contains a radical.
4. Exponents in the radicand and the index of the radical have greatest common factor 1.
5. All indicated operations have been performed (if possible).

Note:

- Numbers such as 1, 4, 9, 16, 25, ... are called **perfect squares**, since $1 = 1^2$, $4 = 2^2$, $9 = 3^2$, $16 = 4^2$, $25 = 5^2$ and so on. Square roots of perfect square radicands simplify to rational numbers.
- Numbers such as 1, 8, 27, 64, 125, ..., are called **perfect cubes**, since $1 = 1^3$, $8 = 2^3$, $27 = 3^3$, $64 = 4^3$, $125 = 5^3$ and so on. Cube roots of perfect cubes radicands simplify to rational numbers.

Example: Simplify $\sqrt{50}$ and $\sqrt[3]{24}$.

$$\sqrt{50} = \sqrt{25 \cdot 2} = \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2}$$

The largest perfect square factor of 50

$$\sqrt[3]{24} = \sqrt[3]{8 \cdot 3} = \sqrt[3]{8} \cdot \sqrt[3]{3} = 2\sqrt[3]{3}$$

The largest perfect cube factor of 24

Problem 3. Simplify each radical. Assume all variable expressions represent positive real numbers.

(a) $\sqrt{288}$

(e) $\sqrt[3]{18x^{10}y^7}$

(b) $\sqrt{4y^{10}}$

(f) $\sqrt[3]{24x^{15}w^{17}}$

(c) $\sqrt{49x^{16}}$

(d) $\sqrt{20s^6t^9}$

(g) $\sqrt[4]{81x^7w^5}$

(h) $\ast \sqrt[5]{243t^{14}u^{16}}$

Problem 4. Add or subtract, as indicated. Assume all variables represent positive real numbers.

(a) $\sqrt{75ab^3} - b\sqrt{12ab}$

(b) $\ast y\sqrt{27yv^2} + 4v\sqrt{12y^3}$

(c) $\ast w\sqrt{3y^3} + 4y\sqrt{75yw^2}$

(d) $\ast x\sqrt{18xu^2} - 6u\sqrt{32x^3}$

Problem 5. Find each product.

(a) $\ast \sqrt{2}(\sqrt{10} + 2\sqrt{3})$

(b) $\ast 6\sqrt{2}(\sqrt{5} - \sqrt{2})$

(c) $\ast \sqrt{5}(\sqrt{7} - 8\sqrt{5})$

(d) $(\sqrt{11} + \sqrt{17})(\sqrt{11} - \sqrt{17})$

(e) $(\sqrt{5} + 1)^2$

$$(f) \text{ * } (\sqrt{x} + 5\sqrt{2})(\sqrt{x} - 5\sqrt{2})$$

$$(g) \text{ * } (\sqrt{x} - \sqrt{5})^2$$

$$(h) \text{ * } (\sqrt{x} + \sqrt{3})(\sqrt{x} - \sqrt{3})$$

$$(i) \text{ * } (\sqrt{x} + 3\sqrt{2})^2$$

Rationalizing Denominators

Condition 3 for a simplified radical requires that no denominator contain a radical. We achieve this by **rationalizing the denominator**—that is, multiplying by a form of 1.

Problem 6. Rationalize each denominator.

$$(a) \frac{2}{\sqrt{5}}$$

$$(c) \text{ * } \sqrt[3]{\frac{4}{9}}$$

$$(e) \text{ * } \sqrt[3]{\frac{6}{5}}$$

$$(b) \text{ * } \sqrt{\frac{3}{10}}$$

$$(d) \text{ * } \sqrt{\frac{6}{5}}$$

$$(f) \text{ * } \frac{5}{\sqrt[3]{4}}$$

Problem 7. Rationalize the denominators of the following expressions.

$$(a) \text{ * } \frac{\sqrt{75} - 20}{30}$$

$$(b) \text{ * } \frac{\sqrt{48} - 6}{10}$$

$$(c) \text{ * } \frac{7}{2\sqrt{5}-1}$$

$$(d) \text{ * } \frac{5\sqrt{3}+2}{\sqrt{3}-1}$$

$$(e) \text{ * } \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$$